

MATH 212

HOMEWORK DUE WEEK 6 MONDAY

Winter 2026

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■ **Question 6001.**

Q6

Prove that the polynomials $\{q_n(x)\}$ where $q_n(x) = \prod_{j=0}^{n-1} (x - x_j)$ are linearly independent.

HINT: use induction.

Solution. For each $n \geq 1$, the polynomial

$$q_n(x) = \prod_{j=0}^{n-1} (x - x_j)$$

is monic of degree n . Indeed, expanding the product shows the leading term is x^n with coefficient 1, regardless of the values of the nodes $x_0, \dots, x_{n-1}$.

Given a linear combination

$$\sum_{n=1}^N c_n q_n(x) \equiv 0.$$

we want to show that $c_1 = \dots = c_N = 0$. We will prove the linear independence of $\{q_n\}_{n=1}^N$ by induction on N .

For $N = 1$, we have $c_1 q_1 \equiv 0$ and since $q_1 \not\equiv 0$, it follows that $c_1 = 0$.

Assume the claim holds for $N - 1$, and consider a linear combination up to N . Consider the coefficient of the highest power x^N in the left-hand side. Since $\deg q_n = n$, no term with $n < N$ contributes to x^N . The only contribution to the x^N coefficient comes from $c_N q_N(x)$, and because q_N is monic, that coefficient is exactly c_N . But the polynomial on the left is identically zero, so its x^N coefficient is 0, hence $c_N = 0$.

With $c_N = 0$, the relation reduces to

$$\sum_{n=1}^{N-1} c_n q_n(x) \equiv 0,$$

and by the induction hypothesis we conclude $c_1 = \dots = c_{N-1} = 0$. Therefore, all coefficients vanish, proving that $\{q_n\}$ is linearly independent. ■

■ **Question 6002.**

□

Let k be a positive integer. Suppose that $p \in \mathcal{P}_{2k+1}$ such that

$$p(-1) = p(1) = 0$$

and

$$p^{(2i-1)}(\pm 1) = 0 \quad \text{for } i = 1, \dots, k$$

Prove that $p \equiv 0$.

Solution. We use Rolle's theorem iteratively.

Step 1: First, $p(-1) = p(1) = 0$ implies by Rolle that p' has at least one zero in $(-1, 1)$. Together with the hypothesis $p'(\pm 1) = 0$, we conclude that p' has at least three distinct zeros in $[-1, 1]$, hence (by applying Rolle on each subinterval between consecutive zeros) p'' has at least two distinct zeros in $(-1, 1)$.

Assume inductively that for some $i \in \{2, \dots, k\}$ the derivative $p^{(2i-2)}$ has at least two distinct zeros in $(-1, 1)$. Applying Rolle to $p^{(2i-2)}$ shows that $p^{(2i-1)}$ has at least one zero in $(-1, 1)$. Since the hypothesis also gives $p^{(2i-1)}(\pm 1) = 0$, the function $p^{(2i-1)}$ has at least three distinct zeros in $[-1, 1]$, and therefore another application of Rolle yields that $p^{(2i)}$ has at least two distinct zeros in $(-1, 1)$.

By induction, $p^{(2k)}$ has at least two distinct zeros in $(-1, 1)$.

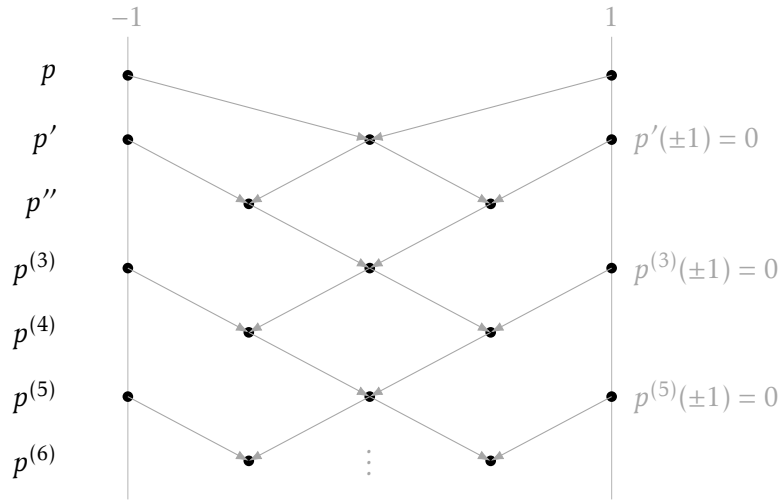


Figure 1: Repeated Rolle argument: odd derivatives have zeros at ± 1 by hypothesis and at least one interior zero by Rolle; even derivatives then have at least two interior zeros.

But $p \in \mathcal{P}_{2k+1}$ implies $p^{(2k)}$ is a polynomial of degree ≤ 1 . A nonzero polynomial of degree ≤ 1 has at most one zero, hence we must have

$$p^{(2k)} \equiv 0.$$

Consequently, $p^{(2k-1)}$ is constant. Since $p^{(2k-1)}(1) = 0$ by hypothesis, it follows that $p^{(2k-1)} \equiv 0$.

Step 2: We now prove by downward induction that $p^{(r)} \equiv 0$ for $r = 2k-1, 2k-2, \dots, 0$.

Suppose $p^{(r+1)} \equiv 0$. Then $p^{(r)}$ is constant.

- If r is odd, then $r \leq 2k-1$ and the hypothesis gives $p^{(r)}(1) = 0$, so the constant $p^{(r)}$ is identically zero.

- If r is even, then $r - 1$ is odd, hence $p^{(r-1)}(-1) = p^{(r-1)}(1) = 0$ by hypothesis. Rolle's theorem applied to $p^{(r-1)}$ yields a point $\xi \in (-1, 1)$ with $p^{(r)}(\xi) = 0$. Since $p^{(r)}$ is constant, it follows that $p^{(r)} \equiv 0$.

This completes the downward induction, and in particular gives $p^{(0)} = p \equiv 0$.



■ Question 6003.

Q15

Note: This question is optional. You do not need to submit an answer.

Suppose that p_n is the polynomial of degree $\leq n$ that interpolates $f(x)$ at the distinct points $x_0, x_1, \dots, x_n$. Develop a conjecture about what p_n converges towards as the interpolating points $x_1, \dots, x_n$ all approach x_0 simultaneously, and explain your thinking.

HINT: use the definition of $E_n(x)$ and the Interpolation Error formula (Cauchy) we developed in class.

Solution. Assume $f \in C^n(I)$ on an interval I containing x_0 . Let $x_0, x_1, \dots, x_n$ be distinct nodes in I , and consider

$$x_j \rightarrow x_0 \quad \text{for } j = 1, \dots, n,$$

while x is fixed.

Claim 0.0.1

The interpolant p_n converges pointwise to the degree- n Taylor polynomial of f at x_0 ,

$$T_n(x) := \sum_{m=0}^n \frac{f^{(m)}(x_0)}{m!} (x - x_0)^m.$$

Heuristic justification via the error term. As the nodes are coalescing,

$$\prod_{j=0}^n (x - x_j) \rightarrow (x - x_0)^{n+1},$$

so the error formula becomes (formally)

$$E_n(x) \approx \frac{f^{(n+1)}(\xi)}{(n+1)!} (x - x_0)^{n+1},$$

which matches the standard form of the Taylor remainder. This suggests that p_n is approaching the unique degree- n polynomial whose error against f is “as small as possible” near x_0 , namely the Taylor polynomial at x_0 .

Proof of claim 1.

Write p_n in Newton form with respect to the ordered nodes:

$$p_n(x) = \sum_{m=0}^n c_m q_m(x), \quad q_m(x) := \prod_{j=0}^{m-1} (x - x_j), \quad q_0 \equiv 1.$$

Let $p_{m-1}(x) := \sum_{\ell=0}^{m-1} c_\ell q_\ell(x)$ be the $(m-1)$ st partial Newton sum, which interpolates f at $x_0, \dots, x_{m-1}$.

Evaluating at x_m and using $p_m(x_m) = f(x_m)$ gives

$$f(x_m) = p_{m-1}(x_m) + c_m q_m(x_m),$$

hence

$$c_m = \frac{f(x_m) - p_{m-1}(x_m)}{\prod_{j=0}^{m-1} (x_m - x_j)}. \quad (1)$$

Apply the Cauchy form of the interpolation error to the degree- $(m-1)$ interpolant p_{m-1} evaluated at $x = x_m$. There exists ξ_m in the convex hull of $\{x_0, \dots, x_m\}$ such that

$$f(x_m) - p_{m-1}(x_m) = \frac{f^{(m)}(\xi_m)}{m!} \prod_{j=0}^{m-1} (x_m - x_j).$$

Substituting into (1) gives the identity

$$c_m = \frac{f^{(m)}(\xi_m)}{m!}.$$

Now send $x_1, \dots, x_m \rightarrow x_0$. Then the convex hull of $\{x_0, \dots, x_m\}$ shrinks to $\{x_0\}$, so $\xi_m \rightarrow x_0$. By continuity of $f^{(m)}$,

$$c_m \rightarrow \frac{f^{(m)}(x_0)}{m!} \quad (m = 0, 1, \dots, n),$$

(with $c_0 = f(x_0)$). Also, for fixed x ,

$$q_m(x) = \prod_{j=0}^{m-1} (x - x_j) \rightarrow (x - x_0)^m.$$

Since the Newton expansion is a finite sum, we may pass to the limit term-by-term:

$$p_n(x) = \sum_{m=0}^n c_m q_m(x) \rightarrow \sum_{m=0}^n \frac{f^{(m)}(x_0)}{m!} (x - x_0)^m = T_n(x).$$

This proves the claim. ■

Conclusion. As the interpolation nodes $x_1, \dots, x_n$ coalesce to x_0 , the interpolating polynomial converges to the degree- n Taylor polynomial of f at x_0 . ■

■ **Question 6004.**

Q21

Suppose S_3 is the natural cubic spline interpolant to $\{(x_j, f_j)\}_{j=0}^n$, and σ is any $C^2[x_0, x_n]$ function that also interpolates the same data. Then show that

$$\|S_3''\|_2 \leq \|\sigma''\|_2.$$

Solution. Let $h := \sigma - S_3$. Then $h \in C^2[x_0, x_n]$ and

$$h(x_j) = \sigma(x_j) - S_3(x_j) = f_j - f_j = 0 \quad (j = 0, \dots, n).$$

Using the L^2 inner product $\langle u, v \rangle := \int_{x_0}^{x_n} u(x)v(x) dx$,

$$\|\sigma''\|_2^2 = \|S_3'' + h''\|_2^2 = \|S_3''\|_2^2 + 2\langle S_3'', h'' \rangle + \|h''\|_2^2.$$

We are going to prove that $\langle S_3'', h'' \rangle = 0$.

Since S_3 is a cubic polynomial on each subinterval $[x_{j-1}, x_j]$, we have $S_3'''' \equiv 0$ on (x_{j-1}, x_j) . On each such interval, integrate by parts twice:

$$\int_{x_{j-1}}^{x_j} S_3'' h'' dx = \left[S_3'' h' \right]_{x_{j-1}}^{x_j} - \int_{x_{j-1}}^{x_j} S_3''' h' dx = \left[S_3'' h' - S_3''' h \right]_{x_{j-1}}^{x_j} + \int_{x_{j-1}}^{x_j} S_3'''' h dx,$$

and the last integral vanishes because $S_3'''' = 0$ on (x_{j-1}, x_j) . Therefore

$$\int_{x_{j-1}}^{x_j} S_3'' h'' dx = \left[S_3'' h' - S_3''' h \right]_{x_{j-1}}^{x_j}.$$

Summing over $j = 1, \dots, n$ gives

$$\langle S_3'', h'' \rangle = \int_{x_0}^{x_n} S_3'' h'' dx = \sum_{j=1}^n \left[S_3'' h' - S_3''' h \right]_{x_{j-1}}^{x_j}.$$

At each interior knot x_j ($1 \leq j \leq n-1$), the terms telescope:

- $h(x_j) = 0$ kills the $S_3''' h$ contributions.
- For $S_3'' h'$, all interior boundary terms cancel. At the endpoints, S_3 is *natural*, so $S_3''(x_0) = S_3''(x_n) = 0$, which kills the remaining endpoint contributions.

Therefore $\langle S_3'', h'' \rangle = 0$.

Consequently,

$$\|\sigma''\|_2^2 = \|S_3''\|_2^2 + \|h''\|_2^2 \geq \|S_3''\|_2^2,$$

and taking square roots yields $\|S_3''\|_2 \leq \|\sigma''\|_2$. ■

■ **Question 6005.**

□

Interpolation connects data pairs using polynomial functions, where the degree of the polynomials is chosen so that all pairs are matched exactly. However, real-world data tend to be inaccurate, in which case exact matching is not necessarily desirable. Instead, we might know (or suspect) that a general type of function (not necessarily polynomial) underlies the data, and we want to choose a particular function of this type to represent the data. Identifying a particular “model” function for the data would allow us not only to predict the behavior of the system at points not represented in the collected dataset, but to study the system more generally. This process of identifying a particular model function for a set of data is called **data fitting** (or “regression”), and we usually choose the model function that “best” fits the data in some sense.

Over the first four weeks, you have seen various linear regression techniques. Recall that the vector $\vec{x}^*$ that minimizes the “Least Square Error” (LSE) $\|A\vec{x} - \vec{b}\|$ is found by solving the Normal equation (you do not need to prove this)

$$A^T A \vec{x}^* = A^T \vec{b}.$$

This exercise will ask you to prove some of the related linear algebra results.

Suppose A is an $m \times n$ matrix (not necessarily square). Prove the following linear algebra results:

(a) $(\text{Col}(A))^\perp = \text{Nul}(A^T).$

Note: $\text{Nul}(A)$ is the nullspace or kernel of A which is defined as $\text{Nul}(A) = \{\vec{v} \in \mathbb{R}^n \mid A\vec{v} = \vec{0}\}.$
 $\text{Col}(A)$ is the column space or range of A which is defined as $\text{Col}(A) = \{A\vec{v} \mid \vec{v} \in \mathbb{R}^n\}.$
 $\text{rank}(A)$ is the dimension of $\text{Col}(A).$

(b) $\text{Nul}(A) = \text{Nul}(A^T A).$

HINT: It is easy to show that $\text{Nul}(A) \subseteq \text{Nul}(A^T A).$ To show the opposite containment $\text{Nul}(A^T A) \subseteq \text{Nul}(A),$ use orthogonal complements.

(c) $\text{rank}(A) = \text{rank}(A^T A).$

Hence if A has full rank, we get $\text{Nul}(A^T A) = \{0\}.$ Consequently, $A^T A$ is invertible, and we have an exact solution to the normal equation $\vec{x}^* = (A^T A)^{-1} A^T \vec{b}.$

Solution. Throughout, we use the standard Euclidean inner product $\langle u, v \rangle = u^T v.$

(a) Let $y \in \mathbb{R}^m.$ Then

$$\begin{aligned} y \in (\text{Col}(A))^\perp &\iff \langle y, Ax \rangle = 0 \text{ for all } x \in \mathbb{R}^n \\ &\iff y^T Ax = 0 \text{ for all } x \\ &\iff (A^T y)^T x = 0 \text{ for all } x \\ &\iff A^T y = 0 \\ &\iff y \in \text{Nul}(A^T). \end{aligned}$$

Thus $(\text{Col}(A))^\perp = \text{Nul}(A^T).$

(b) First, if $x \in \text{Nul}(A)$ then $Ax = 0$, so $A^T Ax = A^T 0 = 0$, hence $\text{Nul}(A) \subseteq \text{Nul}(A^T A).$

For the reverse inclusion, let $x \in \text{Nul}(A^T A)$, i.e. $A^T A x = 0$. Then $Ax \in \text{Nul}(A^T) = (\text{Col}(A))^\perp$ (by part (a)).

But also $Ax \in \text{Col}(A)$ by definition. Therefore

$$Ax \in \text{Col}(A) \cap (\text{Col}(A))^\perp = \{0\},$$

so $Ax = 0$, i.e. $x \in \text{Nul}(A)$. Thus $\text{Nul}(A^T A) \subseteq \text{Nul}(A)$, and we conclude $\text{Nul}(A) = \text{Nul}(A^T A)$.

(c) Both $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $A^T A : \mathbb{R}^n \rightarrow \mathbb{R}^n$ have domain $\mathbb{R}^n$, so Rank–Nullity theorem gives

$$\text{rank}(A) = n - \dim \text{Nul}(A), \quad \text{rank}(A^T A) = n - \dim \text{Nul}(A^T A).$$

By part (2), $\text{Nul}(A) = \text{Nul}(A^T A)$, hence $\text{rank}(A) = \text{rank}(A^T A)$.

Finally, if A has full rank (i.e. $\text{rank}(A) = n$), then $\text{Nul}(A) = \{0\}$, hence $\text{Nul}(A^T A) = \{0\}$ by (2). Since $A^T A$ is an $n \times n$ matrix, trivial nullspace implies invertibility, so the normal equations have the unique solution $x^* = (A^T A)^{-1} A^T b$. ■